• As proved in class / HW / text.

• 2.9 – divegergence through def. of limit via ε–δ & sequences.

• proof techniques, sets, logic.

Math 421

Wednesday, November 5

Announcements

- Grace’s Drop-In Hours
  - **Today** 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
  - Course Assistant: Th 4-7 Table 4
- Homework 8 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
  - Email me & fill out the form if you have a conflict and need to take the makeup.
  - Sign up for McBurney exam if applicable.

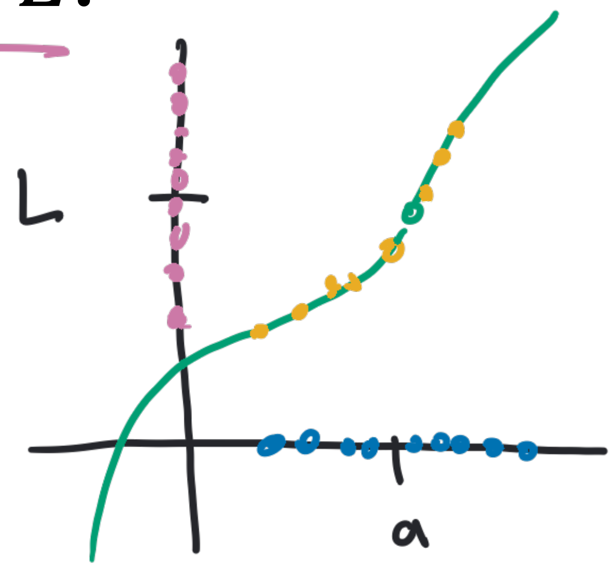
Functional Limits – via Sequences

$S \subseteq \mathbb{R}$

$x_n \in \operatorname{dom} f$.

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S.

Then $\lim_{x\to a} f(x) = L$ if for every sequence $\boxed{(x_n) \text{ in } S}$ satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\underline{\lim x_n = a}$, we have $\underline{\lim f(x_n) = L}$.



Example: Let $f(x) = x^2$, $\lim_{x\to a} x^2 = a^2$

let (x_n) be a sequence, $x_n \in S$, such that $x_n \neq a$ for all $n \in \mathbb{N}$. and $\lim x_n = a$.

$\lim f(x_n) = \lim (x_n)^2 = [\lim x_n]^2 = a^2$

Since (x_n) was arbitary, $\lim_{x\to a} x^2 = a^2$.

Activity

Prove carefully using sequences, and their algebraic limit laws, that

$$\lim_{x \to 3} x^2 - x = 6.$$

Proof. Let (x_n) be a sequence in the domain of $x^2 - x$ such that $x_n \neq 3$ for all $n \in \mathbb{N}$ and $\lim x_n = 3$.

Then $\lim [x_n^2 - x_n] = \lim x_n^2 - \lim x_n =$

$[\lim x_n]^2 - \lim x_n = 3^2 - 3 = 6.$

Since (x_n) was arbitrary, $\lim\limits_{x \to 3} x^2 - x = 6.$

Functional Limits – Two Perspectives

Theorem. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S. Then the following statements are equivalent:

1. $\lim\limits_{x \to a} f(x) = L$
2. for all $\epsilon > 0$ there exists some $\delta > 0$ such that:

   if $0 < |x - a| < \delta$ and $x \in S$, then $|f(x) - L| < \epsilon$.
3. for every sequence (x_n) in S satisfying $x_n \neq a$ for all $n \in \mathbb{N}$ and $\lim x_n = a$, we have $\lim f(x_n) = L$.

for all $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x-a| < \delta$ then $|f(x)-L| < \varepsilon$

Let (x_n) be a seq. $\underbrace{\lim x_n = a}$ Show $\lim f(x_n) = L$.

for all $\cancel{\varepsilon}\ \delta > 0$ there exists N s.t. $n > N$ $|x_n - a| < \cancel{\varepsilon}\ \delta$

Proof of Equivalence

$(\epsilon - \delta \Rightarrow$ Sequences$)$:

Suppose $\lim_{x \to a} f(x) = L$. Let (x_n) be a sequence, $x_n \in S$ and $x_n \neq a$ for all $n \in \mathbb{N}$, and $\lim x_n = a$. Let $\varepsilon > 0$.

Since $\lim_{x \to a} f(x) = L$, there exists $\delta > 0$ such that if $0 < |x-a| < \delta$ then $|f(x) - L| < \varepsilon$. Since $\lim x_n = a$, there exists N such that $n > N$, $|x_n - a| < \delta$. ✱ Since $x_n \neq a$ for all n, $0 < |x_n - a| < \delta$ so $|f(x_n) - L| < \varepsilon$,

since $\varepsilon > 0$ was arbitrary, $\lim f(x_n) = L$.

✱ Assume $n > N$,

Proof of Equivalence

($\epsilon - \delta \Leftarrow$ Sequences):

For every sequence $x_n \in S$, $x_n \neq a$, $\lim x_n = a$, $\lim f(x_n) = L$.

Assume, by way of contradiction, that $\lim_{x \to a} f(x) \neq L$. There exists some $\varepsilon_0 > 0$ such that for all $\delta > 0$ the statement "if $0 < |x-a| < \delta$ then $|f(x) - L| < \varepsilon_0$" is false. In particular it is false for $\delta = \frac{1}{n}$ for all $n \in \mathbb{N}$, so ~~if~~ there exists $x_n \in S$, $0 < |x_n - a| < \frac{1}{n}$, and then $|f(x_n) - L| \geq \varepsilon_0$. Then $-\frac{1}{n} < x_n - a < \frac{1}{n}$ and $a - \frac{1}{n} < x_n < a + \frac{1}{n}$, by the squeeze lemma $\lim x_n = a$, so $\lim f(x_n) = L$, there exists N s.t. $n > N$, $|f(x_n) - L| < \varepsilon_0$, a contradiction.